

UHACKATIK: A NEW 25-KM DIAMETER ~390 Ma IMPACT STRUCTURE IN QUEBEC, CANADA

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In 2024, a Canadian amateur astronomer reported to us a circular structure, centered on Lake Marsal in Quebec (51.2°N, 64.7°W), whose topography was suggestive of an impact structure. Preliminary investigations based on the analysis of the geological maps, of digital elevation data, and of samples provided by the Geological Survey of Canada reinforced the idea that the structure was potentially a hypervelocity impact structure [1].

An expedition to the structure was organized in October 2025 to perform field observations and sampling. In the central part of the structure, we found decimeter-sized shatter cones in the Grenvillian basement (Fig. 1A). We reinterpreted the Marsal Breccia Formation – outcropping between 2 and 4 km away from the center and initially interpreted as a diatreme formation [2] – as a clast-poor impact melt rock. This formation is up to 50 m-thick and displays characteristic columnar jointing (Fig. 1B). It contains basement clasts up to 50 cm in diameter. Oriented cores were collected in this formation for paleomagnetism. The paleomagnetic direction indicate a formation age of ~390 Ma (370–400 Ma), during a reverse polarity interval of the geomagnetic field. This age is corroborated by preliminary data obtained on zircons in the impact melt rocks, whose LA-ICPMS lead isotopic data point toward a youngest age near 350–400 Ma.

Our data thus confirm that this structure is of hypervelocity impact origin, with a diameter of ~25 km formed at ~390 Ma on the mafic Grenvillian basement. The name of Uhackatik was chosen following discussions with the Ekuanitshit Innu council. This structure is remarkable for the relatively good preservation of a ≥50 m-thick impact melt rock sheet and the overall preservation of a clear topographic signal despite the structure age. It is also noteworthy that an impact structure 25 km in diameter can still be discovered by the general public based on available satellite imagery.

References: [1] Rochette et al. (2024) Lake Marsal: a large impact structure candidate in Quebec, Canada. 86th meeting of the Meteoritical Society, abstract #6009 [2] Chevé et al. (2001). Géologie de la région de la rivière Baune (22P/02) et des lacs à l'Aigle (22P/03) et Canatiche (22P/04). Géologie Québec, 58pp.

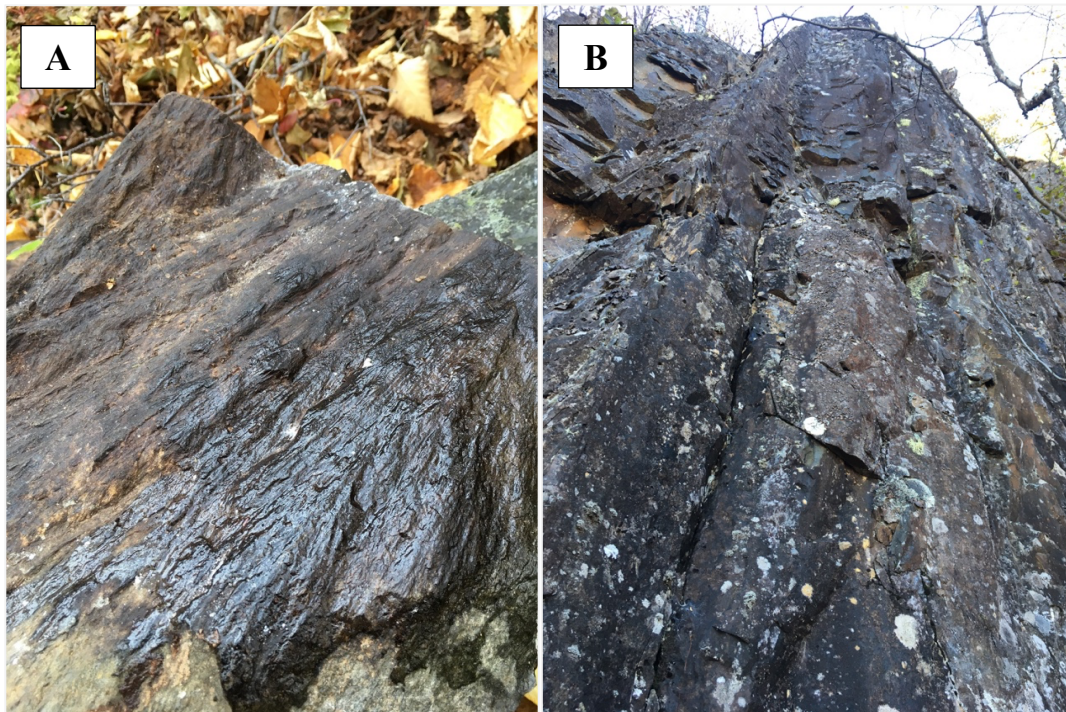


Figure 1. A) Shatter cone in the Grenvillian basement at the center of the structure. The block is 20 cm in size. B) 25m-high cliff of impact melt rock displaying columnar jointing, 4 km west of the structure center.